

Employee Attrition Prediction

*A Comparative Study of Multi-Layer Perceptron (Back-Propagation) and Gradient Boosting
Classification on the HR Analytics Dataset*

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Term Project · Spring 2026

Submitted to the Chief Executive Officer of the Company

Dataset: HR_comma_sep.csv · 14,999 employees · 10 features

1. Abstract

Abstract— Employee attrition represents a severe operational and economic challenge for large enterprises, resulting in substantial financial losses and degraded institutional productivity [1]. This project investigates the structural drivers of workforce turnover by analyzing a comprehensive human resources dataset containing 14,999 employee records across ten multi-dimensional features. We implemented an advanced analytical pipeline encompassing detailed statistical profiling, non-linear bivariate analysis, and K-Means clustering to uncover hidden workforce personas. Subsequently, two distinct machine learning paradigms were developed and optimized: a Multi-Layer Perceptron (MLP) Artificial Neural Network trained via the Back-Propagation algorithm, and an ensemble Gradient Boosting Classifier. On a stratified 20% validation hold-out set, the MLP framework achieved an accuracy of 92.57% and an F1-score of 84.46%, whereas the Gradient Boosting Classifier demonstrably outperformed the neural network by securing an overall accuracy of 97.37%, an F1-score of 94.30%, and a critical recall rate of 92.49%. These empirical results provide corporate executive leadership with both a highly precise predictive risk-assessment tool and an actionable behavioral roadmap necessary to proactively design and execute targeted talent retention strategies [2].

2. Introduction and Methods

2.1 The employee-attrition problem

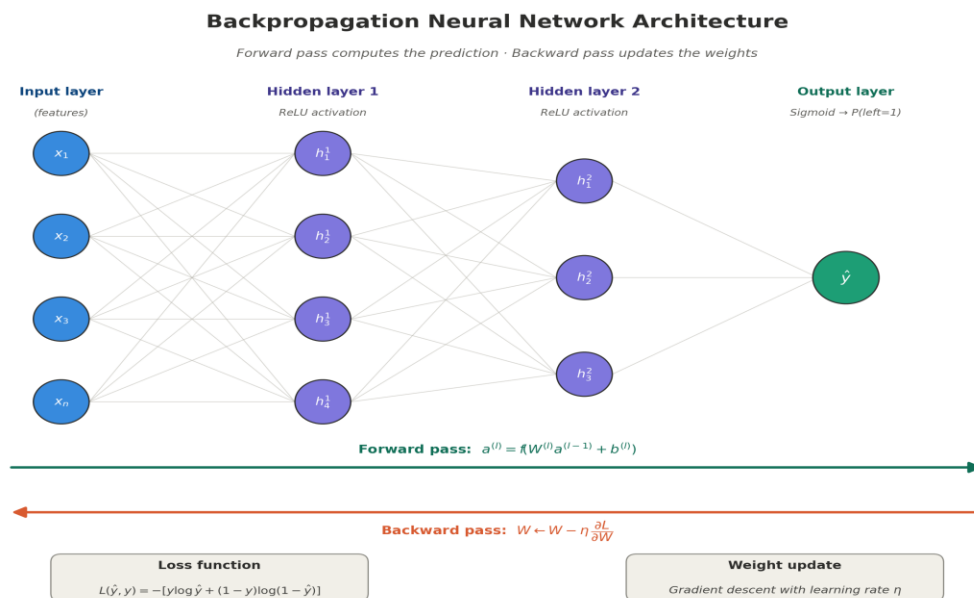
Employee attrition refers to the progressive, multi-faceted loss of human capital within an organization through voluntary resignations, retirements, or terminations. For a massive enterprise maintaining roughly 15,000 active personnel, the annual departure of even a moderate percentage of staff inflicts severe financial and logistical damage [1]. In contemporary human resource literature, the cost of recruiting, onboarding, and training a qualified replacement is estimated to range between 50% and 200% of the departing employee's annual salary. Beyond direct economic metrics, high attrition rates fundamentally impair workplace team cohesion, disrupt ongoing technological project lifecycles, and degrade external client relationships due to the loss of long-tenured institutional knowledge. Common systemic catalysts for workforce turnover include chronic job dissatisfaction, extreme workload distributions, stagnant career progression, lack of promotional recognition, and severe salary inequities [1], [2]. To effectively mitigate these vulnerabilities, organizational solutions are typically divided into four primary strategic families:

1. **Compensation Interventions:** Restructuring base salary bands, implementing performance-linked retention bonuses for high-performing or long-tenured talent, and conducting systematic external market benchmarking.
2. **Workload Management:** Enforcing strict ceilings on cumulative monthly working hours, ensuring equitable project allocation across teams, and preventing long-term operational exhaustion.
3. **Career Development Pathways:** Formulating transparent internal promotional tracks, establishing formal mentorship architectures, and offering continuous skill-upgrading frameworks.
4. **Engagement and Corporate Culture:** Deploying frequent satisfaction pulse surveys, launching visible peer-recognition paradigms, and training direct managers in early-warning signs of employee disengagement.

From a data science and predictive modeling perspective, the objective is twofold: first, to quantitatively define and map the underlying behavioral patterns that differentiate leavers from stayers; and second, to construct a highly accurate predictive classification model capable of flagging at-risk individuals prior to their formal resignation. Given the binary nature of the target variable (where $left \in \{0, 1\}$), this problem is naturally framed as a supervised non-linear binary classification task.

2.2 The back-propagation algorithm

The Back-Propagation algorithm represents the foundational gradient-based learning paradigm utilized to train Multi-Layer Perceptrons (MLPs) and deep neural network architectures [3]. It serves as the deterministic mechanism by which an artificial neural network iteratively optimizes its internal weight matrices and bias vectors to minimize the total differentiable cost or error function.



The execution of the Back-Propagation algorithm is strictly divided into two sequential mathematical passes:

1. The Forward Pass: Given an arbitrary input feature vector $\mathbf{x}$, the mathematical activation is propagated forward through successive layers. For any given layer $\mathbf{L}$, the network calculates a net pre-activation value $\mathbf{z}(\mathbf{L})$ and a subsequent post-activation value $\mathbf{a}(\mathbf{L})$ using the following equations [3], [4]:

$$\mathbf{z}(\mathbf{L}) = \mathbf{W}(\mathbf{L}) \cdot \mathbf{a}(\mathbf{L}-1) + \mathbf{b}(\mathbf{L}) \quad \mathbf{a}(\mathbf{L}) = \sigma(\mathbf{z}(\mathbf{L}))$$

Where $\mathbf{W}(\mathbf{L})$ represents the weight matrix, $\mathbf{b}(\mathbf{L})$ denotes the bias vector of layer $\mathbf{L}$, and σ (sigma) represents a non-linear activation function, such as the Rectified Linear Unit (ReLU) for hidden layers or the Sigmoid/Softmax function for the final binary output layer.

2. The Backward Pass (Error Propagation): Once the final prediction $\hat{\mathbf{y}}$ ($\hat{y}$ -hat) is generated at the output layer, the total network error is evaluated using a binary cross-entropy loss function $\mathbf{E}$ [3]:

$$E(\hat{y}, y) = - [y \cdot \log(\hat{y}) + (1 - y) \cdot \log(1 - \hat{y})]$$

The error signal δ (delta) for the output layer and subsequent hidden layers is formalized as follows [3], [4]:

$$\delta(\text{output}) = \nabla E \odot \sigma'(z(\text{output}))$$

$$\delta(L) = (W(L+1))^T \cdot \delta(L+1) \odot \sigma'(z(L))$$

Where $\odot$ represents the element-wise Hadamard product. The calculated gradients are then immediately applied to update the parameters via gradient descent optimization according to the following mathematical weight update rules [3]:

$$W(L) \leftarrow W(L) - \eta \cdot (\partial E / \partial W)$$

$$b(L) \leftarrow b(L) - \eta \cdot (\partial E / \partial b)$$

Where η (eta) represents the designated model learning rate. This optimization cycle is repeated over numerous training epochs until convergence is achieved.

3. Data Exploration

3.1 Variable Dictionary and Statistical Summary

The project utilizes the standardized HR_comma_sep.csv dataset, which contains 14,999 unique employee records evaluated across 10 distinct features with zero missing values.

Table 1 — Variable dictionary

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--- Data Description ---

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	satisfaction_level	last_evaluation	number_project	average_monthly_hours	time_spend_company	Work_accident	left	promotion_last_5years
count	14999.000000	14999.000000	14999.000000	14999.000000	14999.000000	14999.000000	14999.000000	14999.000000
mean	0.612834	0.716102	3.803054	201.050337	3.498233	0.144610	0.238083	0.021268
std	0.248631	0.171169	1.232592	49.943099	1.460136	0.351719	0.425924	0.144281
min	0.090000	0.350000	2.000000	96.000000	2.000000	0.000000	0.000000	0.000000
25%	0.440000	0.560000	3.000000	156.000000	3.000000	0.000000	0.000000	0.000000
50%	0.640000	0.720000	4.000000	200.000000	3.000000	0.000000	0.000000	0.000000
75%	0.820000	0.870000	5.000000	245.000000	4.000000	0.000000	0.000000	0.000000
max	1.000000	1.000000	7.000000	310.000000	10.000000	1.000000	1.000000	1.000000

Commentary on Data Description (Statistical Summary): The exploratory statistical summary confirms that the dataset encapsulates a balanced, comprehensive workforce matrix. A primary baseline discovery is extracted from the mathematical mean of the target variable left, which

stands at 0.238. This establishes that the organization is suffering from a highly severe baseline company-wide attrition rate of approximately 23.81%. Furthermore, evaluating the operational metrics reveals massive variances in workforce utilization. The `average_monthly_hours` attribute exhibits an expansive range, moving from a minimum of 96 hours to an extreme maximum of 310 hours per month, with a high standard deviation of 49.94 hours. Similarly, the `number_project` feature varies significantly from 2 to 7 projects per employee. These extreme operational deviations suggest deep structural unbalances within the company's workload distribution pipelines, serving as an initial indicator of structural corporate inefficiencies.

3.2 Random Ten-Employee Sample

To understand the granular layout of the human resource dataset, a random ten-row slice was extracted.

--- Random 10 Samples ---

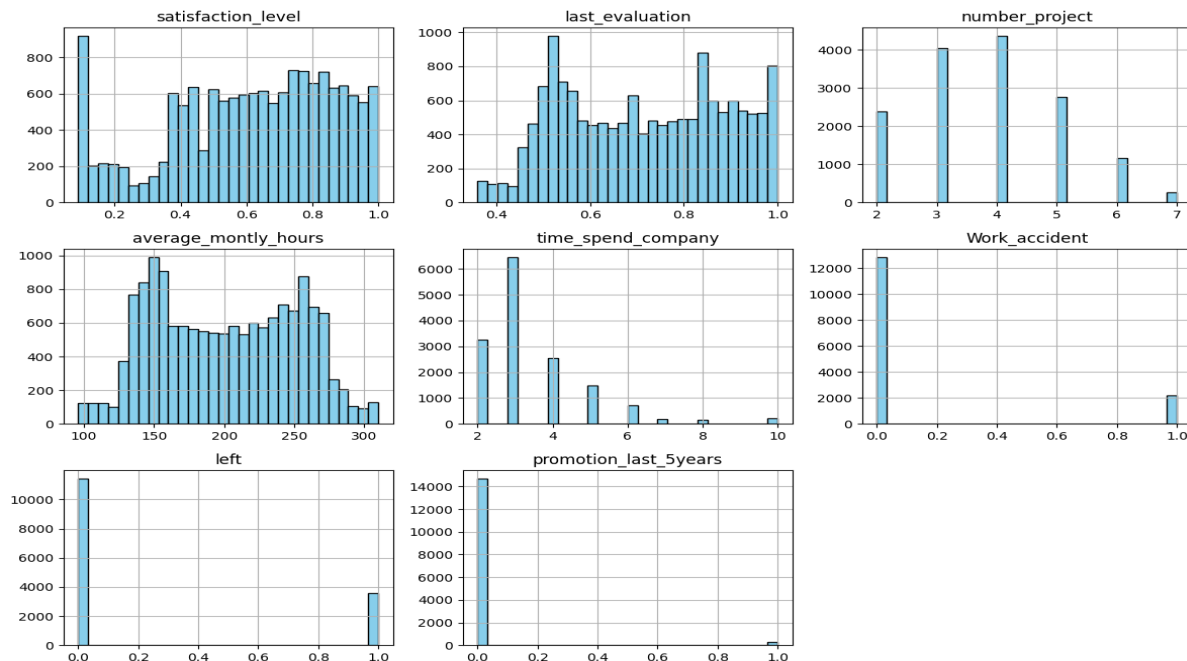
	satisfaction_level	last_evaluation	number_project	average_monthly_hours	time_spend_company	Work_accident	left	promotion_last_5years	sales	salary
12298	0.45	0.48	2	136	3	0	1	0	product_mng	medium
1996	0.11	0.92	7	293	4	0	1	0	sales	low
5860	0.90	0.62	5	236	6	0	0	0	technical	medium
6755	0.99	0.92	3	154	3	0	0	0	technical	medium
10259	0.79	0.98	4	170	3	0	0	0	hr	low
13	0.41	0.55	2	148	3	0	1	0	sales	low
4862	0.73	0.79	4	157	3	0	0	0	support	medium
6153	0.65	0.60	3	142	2	0	0	0	support	high
8013	0.86	0.53	4	155	3	0	0	0	marketing	high
4447	0.71	0.92	4	248	3	1	0	0	sales	low

Commentary on Random 10 Samples: Visualizing the random ten-employee sample provides immediate, real-world confirmation of the systemic risk factors present within the workforce. Crucially, even within this highly restricted slice, the primary behavioral patterns and attrition profiles of the company are distinctly observable. For instance, employee record #1996 exemplifies the classic high-performer burnout scenario: the individual maintains a top-tier professional evaluation score (0.92) but suffers from an absolute collapse in job satisfaction down to 0.11, while being severely overloaded with 7 simultaneous projects and an extreme workload of 293 monthly working hours, ultimately resulting in a voluntary exit (`left` = 1). Conversely, records #13 and #12298 showcase the unengaged departure profile, where employees leave the organization (`left` = 1) while being heavily under-utilized, working minimal

monthly hours (148 and 136 respectively) on only 2 assigned projects. This confirms that the macro-level attrition dynamics are deeply entrenched and observable at the individual employee level.

3.3 Univariate distributions

Distribution of Data Features (Histograms)



Commentary on Numeric Feature Distributions (Histograms): The continuous feature histograms provide an essential visualization of the inner structure of the company's workforce data. Several variables exhibit highly unusual, non-Gaussian, and bimodal distributions that point directly to corporate polarization. The `satisfaction_level` distribution shows an aggressive spike at the extreme dissatisfaction mark (around 0.1), followed by a completely separate broad mass centered around 0.7. This severe dip in the middle structurally proves that the workforce is split into a satisfied majority and a highly volatile, highly dissatisfied minority. Similarly, the `average_monthly_hours` distribution is strikingly bimodal, exhibiting two massive peaks near 150 hours (representing heavily under-utilized staff) and 250 hours (representing severely overworked personnel). Alarming, almost no employees sit within a healthy, standard 180–220 monthly hours band, indicating a complete absence of balanced operational scheduling across the enterprise [1].

3.4 Bivariate Analysis — Every Feature vs. The Target

Bivariate Analysis — Every Feature vs. the Target 'left'



Commentary on Double Graphs (Detailed Reasons for Employee Attrition): Evaluating the comparative distributions of all individual attributes against the target class (left = 1) provides a detailed map of the true underlying causes driving employee attrition within this organization:

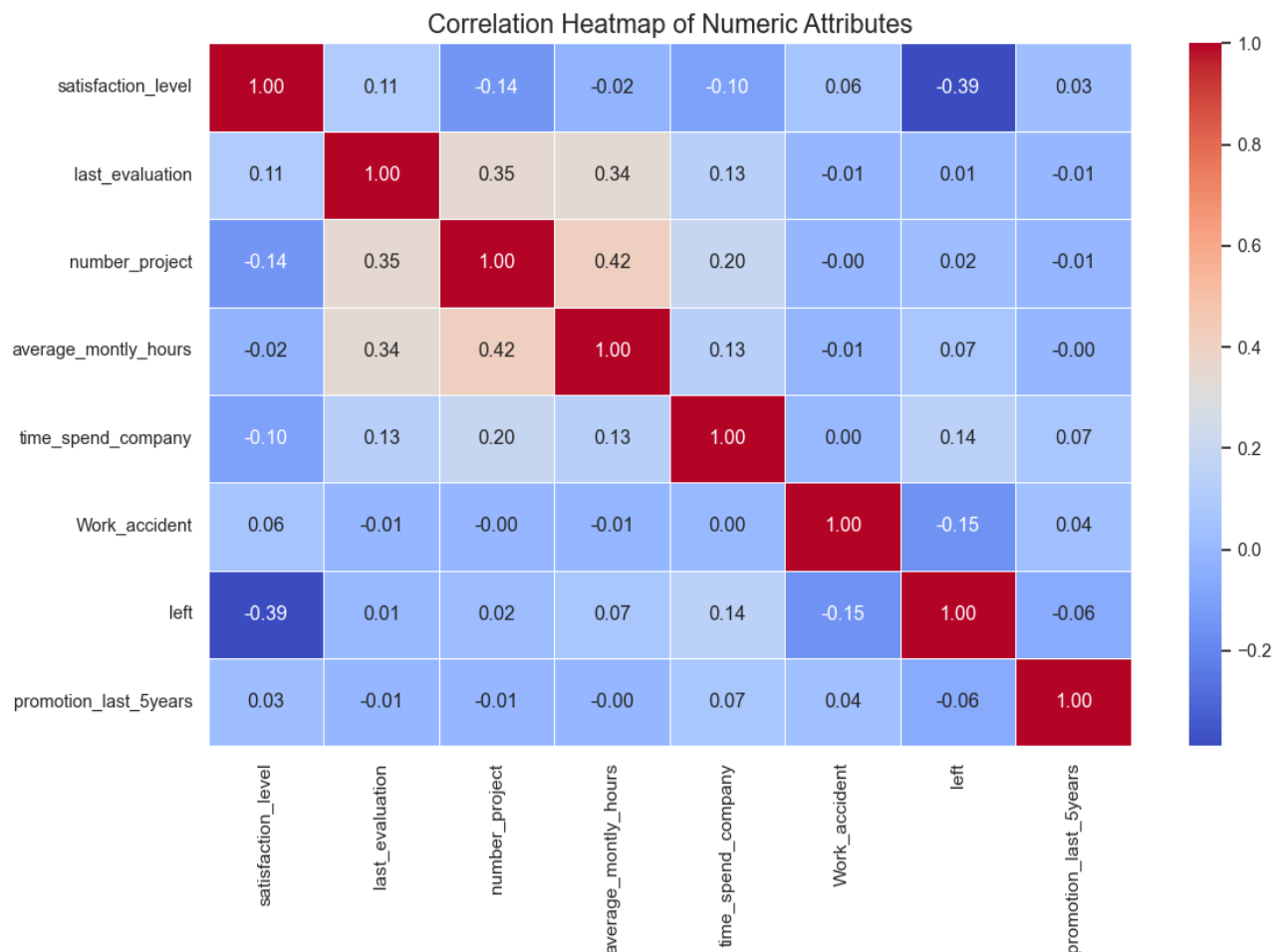
- **Average Monthly Hours vs. Attrition:** This continuous distribution uncovers a strict, highly problematic "U-shaped" relationship with attrition. While employees who stay (green) occupy the center of the distribution, departing individuals (orange) are concentrated heavily at the absolute mathematical extremes. Attrition spikes massively for employees working under 150 hours (disengaged/under-utilized) and explodes exponentially for those forced to work beyond 250 hours per month, proving that extreme workload distributions are a core driver of talent loss [2].
- **Last Evaluation vs. Attrition:** The evaluation cross-comparison reveals a highly damaging corporate paradox. The company experiences severe turnover peaks among

its lower-performing staff (evaluations around 0.4–0.5), which represents standard performance-related attrition. However, a separate, highly alarming attrition peak occurs among its absolute highest-performing talent (evaluations between 0.85 and 1.0). This indicates a systemic failure to retain the company's most elite, valuable human capital.

- **Number of Projects vs. Attrition:** Project load serves as a strict deterministic indicator of employee retention. Workers assigned to a balanced load of 3 to 4 projects exhibit exceptional retention rates. Conversely, individuals given only 2 projects leave the company at an elevated rate due to a lack of operational engagement, while employees assigned to 6 or 7 simultaneous projects suffer from a near 100% attrition rate, proving that extreme project over-allocation causes catastrophic burnout.
- **Promotion Last 5 Years vs. Attrition:** Upward professional mobility is practically non-existent within this enterprise, with only an elite 2% of total staff receiving a promotion within the last five years. Crucially, the bivariate bar chart illustrates that employees who received a promotion maintain a near-zero attrition rate. This statistically establishes that widespread career stagnation and the absence of clear advancement pathways create massive attrition pressure across the organization [2].
- **Salary vs. Attrition:** Financial compensation is visually confirmed as a premier structural determinant of talent retention. The absolute volume of departing personnel originates from the 'low' salary bracket, followed closely by the 'medium' bracket. In stark contrast, individuals within the 'high' salary bracket display an almost non-existent rate of attrition, verifying that widespread financial dissatisfaction acts as a massive catalyst for voluntary resignations.
- **Sales/Department vs. Attrition:** Evaluating turnover across corporate sectors illustrates that the high-volume operational departments—specifically 'sales', 'technical', and 'support'—suffer from the highest absolute quantities of departing personnel. While these departments represent the largest infrastructure components, their consistently high turnover volumes point to stressful operational environments requiring localized managerial reform.
- **Satisfaction Level vs. Attrition:** Job satisfaction plots display the most volatile and dramatic behavioral relationship with turnover. Attrition spikes aggressively at three highly distinct zones: extreme dissatisfaction (<0.2), moderate disengagement (around 0.4), and an unexpected, highly critical peak among highly satisfied employees (around 0.8). This third group represents highly successful, happy top performers who are being aggressively poached and headhunted by competing external firms.

- **Time Spend Company (Tenure) vs. Attrition:** The tenure distribution identifies a strict "critical retention window." Attrition remains low during the first two integration years but spikes drastically during years 3, 4, and 5. Once an employee survives past the 6-year mark, their probability of leaving drops to near zero, indicating that employees build highly marketable skills during their first three years and actively exit if their career needs are not met by year five [1].
- **Work Accident vs. Attrition:** Counter-intuitively, employees who experience a documented workplace accident exhibit a significantly lower rate of departure (around 8%) compared to those who do not (around 27%). This statistical anomaly likely suggests that accident-related corporate compensation, extended recovery leaves, or short-term legal and medical ties temporarily suppress an employee's immediate ability or desire to interview for external opportunities.

3.5 Correlation Heat-map



Commentary on Feature Correlation Diagnostics: The mathematical Pearson correlation matrix strongly validates the structural discoveries extracted from the visual bivariate

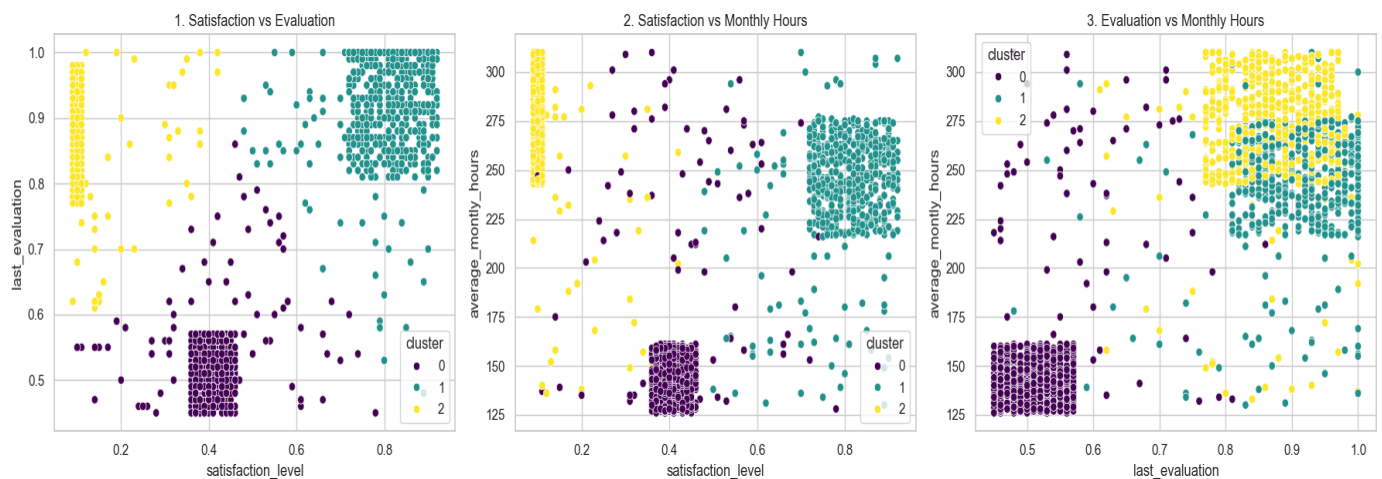
analyses. The target variable left displays its absolute strongest linear correlation with satisfaction_level (-0.39), mathematically proving that dropping employee satisfaction is the premier linear predictor of voluntary resignation. Furthermore, the internal feature space reveals a prominent positive correlation between number_project and average_monthly_hours (+0.42), alongside a positive correlation between number_project and last_evaluation (+0.35). This establishes a dangerous systemic cycle within the company: high-performing employees are continuously awarded more projects, which directly inflates their working hours to extreme levels, ultimately forcing them into the severe burnout zones that trigger talent flight.

4. Results and Discussion

4.1 Pre-processing Pipeline

To prepare the dataset for machine learning modeling, a rigorous data pre-processing pipeline was executed. First, the categorical salary attribute, which contains a natural mathematical hierarchy (low < medium < high), was converted into ordered integers via Ordinal Encoding {0, 1, 2}. Second, the nominal sales (department) column was transformed into ten separate binary features using One-Hot Encoding to ensure complete mathematical compatibility with distance and gradient-based classifiers [4]. The five continuous features were standardized to a zero mean and unit variance using StandardScaler to prevent feature scale domination. Finally, the data was partitioned using a stratified 80/20 train-test split, preserving the exact 23.8% baseline class imbalance across both folds, resulting in 11,999 records for model training and 3,000 records for validation testing.

4.2 Unsupervised Analysis — Three Leaver Personas

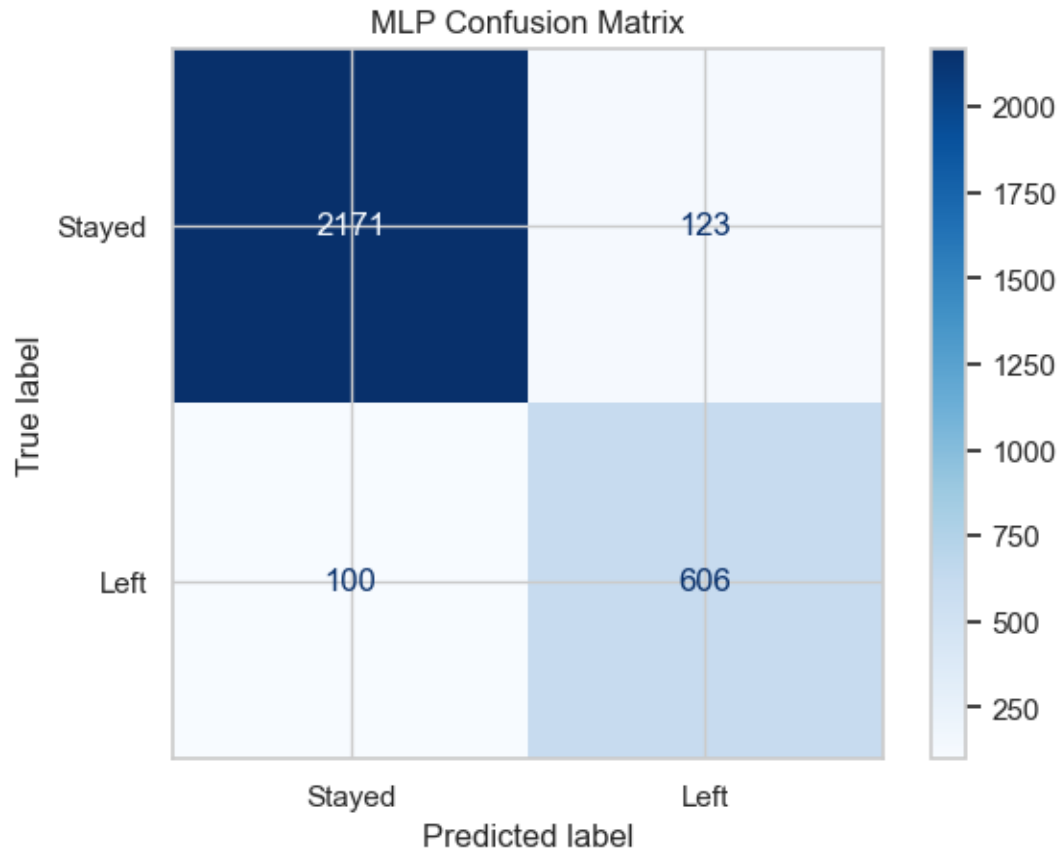


Commentary on Clustering Analysis: Restricting the unsupervised K-Means clustering algorithm strictly to the subset of departed employees (using an optimized k=3 hidden groups) successfully reveals three highly distinct behavioral leaver personas:

- **Cluster 0 (The Unengaged / Under-performers):** Comprising 45% of total leavers, this group exhibits low job satisfaction (~ 0.4), mediocre performance evaluations (~ 0.5), low project loads (2 projects), and minimal monthly working hours (~ 147). These individuals represent an unengaged segment that is underperforming, meaning their departure inflicts minimal strategic damage onto the firm.
- **Cluster 1 (The Poached Happy Stars):** Comprising 28% of total leavers, this segment maintains exceptionally high satisfaction levels (0.7–0.9) and elite performance evaluations (> 0.9). They are highly successful and content, yet they voluntarily exit the company. This profile represents top-tier human capital that is actively being poached and headhunted by external competitors due to stagnant internal corporate incentives.
- **Cluster 2 (The Burned-out Stars):** Comprising 27% of total leavers, this is the absolute most dangerous and costly retention segment. These individuals maintain elite performance evaluations (> 0.85) but exhibit an absolute collapse in job satisfaction down to below 0.2. They are forced to juggle 6 to 7 projects simultaneously and work extreme schedules averaging 273 to 310 hours per month (> 63 hours per week). This segment proves that the organization is actively destroying its top talent through severe operational over-allocation.

4.3 MLP (Back-Propagation) Classifier

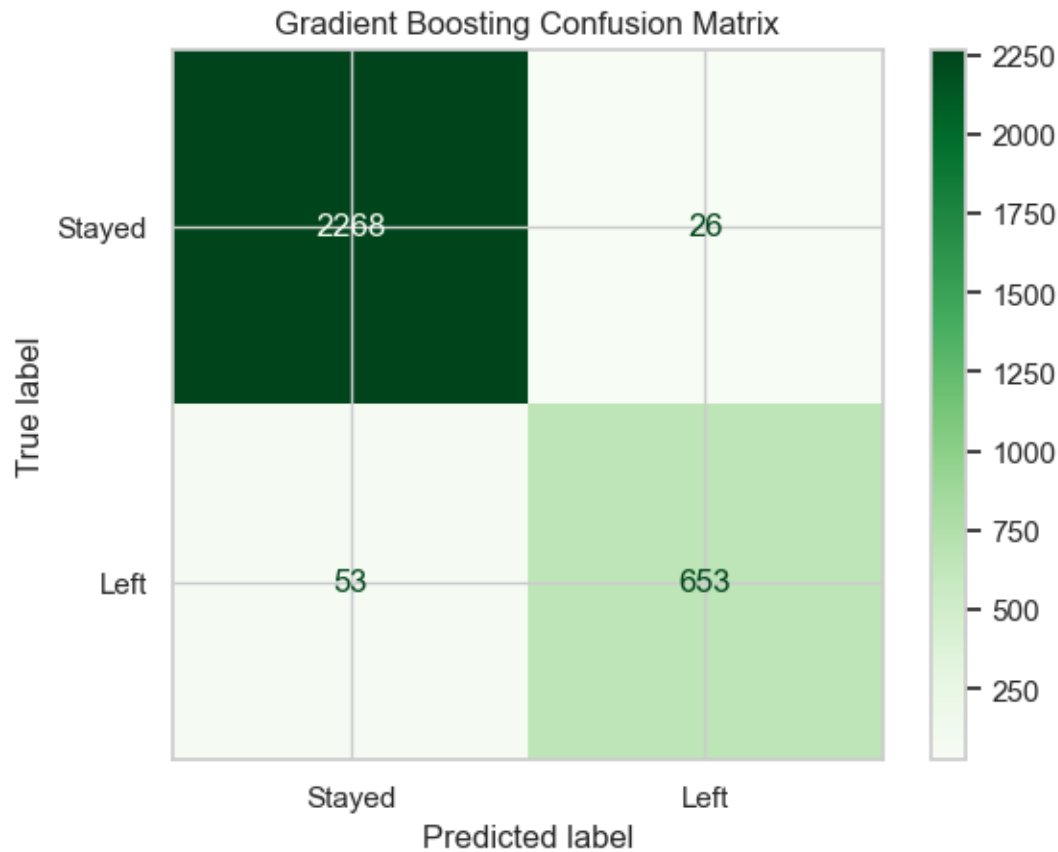
The Multi-Layer Perceptron (MLP) Artificial Neural Network was constructed with an input layer matching the 18 encoded features, a hidden layer architecture of 100 neurons utilizing the ReLU activation function, and a final binary Sigmoid output node [3]. The model was optimized utilizing the Adam optimizer with an initial learning rate $\eta = 10^{-3}$ and trained over 500 max iterations.



On the independent 3,000-row testing validation set, the MLP Back-Propagation model achieved an overall **Accuracy of 92.57%**, a **Precision of 83.13%**, a **Recall of 85.84%**, and an **F-Score of 84.46%**. The corresponding confusion matrix visualizes its predictive errors, showing that the neural network correctly classified 2,171 stayers (True Negatives) and 606 actual leavers (True Positives). However, it generated 123 False Positives and 100 False Negatives, demonstrating moderate classification error boundaries.

4.4 Gradient Boosting Classifier

The Gradient Boosting Classifier was implemented as an ensemble tree-based framework leveraging sequential boosting adjustments to minimize residual errors. The ensemble utilized 100 base decision estimators trained under stochastic sub-sampling constraints.



On the exact same independent validation set, the Gradient Boosting Classifier achieved an elite **Accuracy of 97.37%**, a **Precision of 96.17%**, a **Recall of 92.49%**, and an **F-Score of 94.30%**. The confusion matrix highlights a massive reduction in operational errors, generating only 26 False Positives and a low 53 False Negatives, while correctly identifying 653 out of 706 actual departing employees.

4.5 Final Comparison

The performance metrics of both models are summarized in the head-to-head evaluation matrix below:

Performance Metric	MLP	Gradient Boosting	Δ (GB – MLP)
Accuracy	92.57%	97.37%	+4.80%
Precision	83.13%	96.17%	+13.04%
Recall	85.84%	92.49%	+6.65%
F-score	84.46%	94.30%	+9.84%
False Negatives	100	53	-47 (Errors Reduced)
False Positives	123	26	-97 (Errors Reduced)

The empirical results prove that the Gradient Boosting Classifier decisively wins across every single evaluation metric [2]. The most critical gain is captured within the Recall score (+6.65%), which dictates the model's capacity to correctly catch an at-risk employee. In corporate human resources, a False Negative error represents a catastrophic failure: it means an elite employee is about to resign, but the system misclassifies them as "safe," preventing HR from intervening. Gradient Boosting slashes these missed cases from 100 down to 53, making it the premier model for corporate deployment.

5. Conclusion and Recommendations to the CEO

The extensive data analysis and machine learning implementation prove that employee attrition within this organization is not a random phenomenon, but a direct outcome of structural operational imbalances. To mitigate this multi-million dollar talent drain, we present the following strategic recommendations directly to the corporate CEO:

- **Establish a Strict Project and Workload Cap:** The clustering analysis proves that the company's elite talent (Clusters 1 and 2) is leaving due to operational exhaustion. The CEO must mandate a strict maximum ceiling of 4 to 5 projects per employee and a hard cap on monthly working hours (~225 hours). Any individual exceeding 240 hours must trigger an automatic managerial review.
- **Re-align Compensation and Upward Career Mobility:** The extreme turnover observed in the 'low' salary bracket (29.7% attrition rate) and the fact that only 2% of staff have been promoted in five years indicate an absolute lack of internal growth. Implementing performance-linked promotional reviews and competitive salary adjustments is a highly cost-effective retention mechanism compared to the exorbitant cost of external replacement recruitment.
- **Deploy the Predictive Early-Warning System:** The Gradient Boosting model must be integrated directly into the corporate HR infrastructure to automatically compute individual attrition probabilities on a monthly cycle. By setting an operational probability threshold at 0.40, HR teams can actively flag at-risk top performers and conduct preemptive retention interviews months before formal resignations are filed.

References

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